

MATHEMATICS 2210
Final, Fall semester 2012

Name: Key

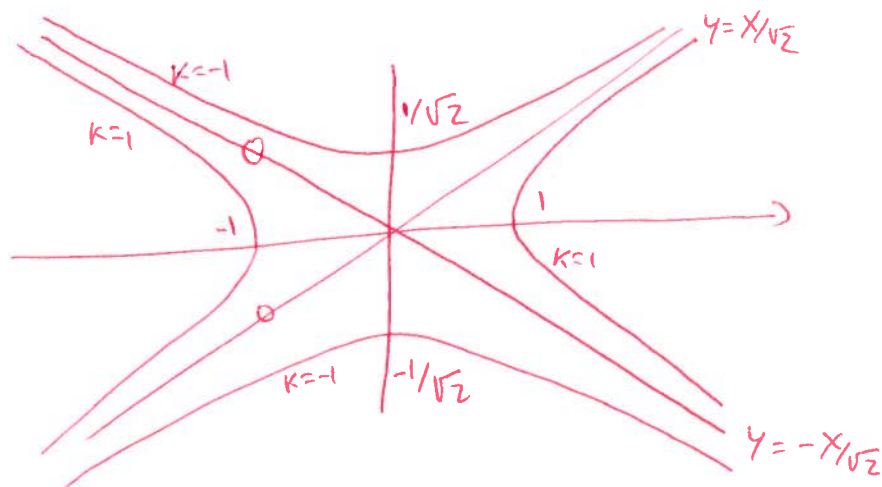
1. (10 pts) Find the arc length of $\mathbf{r}(t) = (\frac{1}{2}t^2, \frac{1}{3}t^3, \frac{1}{3}t^3)$ where $1 \leq t \leq 3$.

$$\mathbf{r}'(t) = (t, t^2, t^2) \quad \|\mathbf{r}'(t)\| = \sqrt{t^2 + t^4 + t^4} = \sqrt{t^2(1+2t^2)} = t\sqrt{1+2t^2}$$

$$\int_1^3 \|\mathbf{r}'(t)\| dt = \int_1^3 t\sqrt{1+2t^2} dt \quad u = 1+2t^2 \quad du = 4t dt$$

$$= \int \frac{1}{4} u^{1/2} du = \left[\frac{1}{6} u^{3/2} \right] = \left[\frac{(1+2t^2)^{3/2}}{6} \right]_1^3 = \frac{19^{3/2} - 3^{3/2}}{6}$$

2. (10 pts) Sketch the level curves of $f(x, y) = x^2 - 2y^2$ for $k = -1, 0, 1$.



$$\begin{aligned} x^2 - 2y^2 &= -1 \\ x^2 - 2y^2 &= 0 \\ x^2 - 2y^2 &= 1 \end{aligned}$$

3. (10 pts) Find the critical points of $f(x, y) = x^2 + y^2 + x^2y + 4$. Which ones are maxima, minima or saddles?

$$\nabla f = (2x + 2xy, 2y + x^2) \quad \begin{cases} 2x + 2xy = 0 & 2x - x^3 = 0 \rightarrow x(2 - x^2) = 0 \\ 2y + x^2 = 0 & \rightarrow 2y = -x^2 \end{cases} \quad \begin{matrix} \uparrow \\ x = 0, \pm\sqrt{2} \end{matrix}$$

$$f_{xx} = 2 + 2y \quad f_{xy} = 2x \quad f_{yy} = 2$$

$$D = (2 + 2y) \cdot 2 - 4x^2$$

critical pts	D
(0, 0)	$4 > 0 \quad f_{xx}(0, 0) = 2 > 0 \quad \underline{\text{MIN}}$
$(\sqrt{2}, -1)$	$\begin{matrix} -8 < 0 \\ -8 < 0 \end{matrix} \quad \text{saddle}$
$(-\sqrt{2}, -1)$	

4. (10 pts) Find the maxima and minima of $f(x, y, z) = 2x + y - 3z$ subject to the constraint $2x^2 + y^2 + 2z^2 = 1$.

$$\begin{cases} \nabla f = \lambda \nabla g \\ g = 0 \end{cases} \Rightarrow \begin{cases} 2 = \lambda 4x \\ 1 = \lambda 2y \\ -3 = \lambda 4z \end{cases}$$

$$2x^2 + y^2 + 2z^2 = 1$$

$$\frac{1}{\lambda} = 2x = 2y = -\frac{4}{3}z$$

$$y = x \quad z = -\frac{3}{2}x$$

$$2x^2 + x^2 + \frac{9}{2}x^2 = 1$$

$$\frac{15}{2}x^2 = 1 \quad x = \pm \sqrt{\frac{2}{15}}$$

$$f\left(\sqrt{\frac{2}{15}}, \sqrt{\frac{2}{15}}, -\frac{3}{2}\sqrt{\frac{2}{15}}\right) = \sqrt{\frac{2}{15}} \cdot \left(2 + 1 + \frac{9}{2}\right)$$

$$= \sqrt{\frac{2}{15}} \cdot \frac{15}{2} = \sqrt{\frac{15}{2}} \quad \text{MAX}$$

$$f\left(-\sqrt{\frac{2}{15}}, -\sqrt{\frac{2}{15}}, \frac{3}{2}\sqrt{\frac{2}{15}}\right) = -\sqrt{\frac{15}{2}} \quad \text{MIN}$$

5. (15 pts) Compute the work done by the force field $F(x, y, z) = x^2 \mathbf{i} + y \mathbf{j} + y \mathbf{k}$ in moving a particle along the curve $\mathbf{r}(t) = (t, \sin(t), \cos(t))$ where $-\pi \leq t \leq \pi$.

$$W = \int F \cdot d\mathbf{r} = \int_{-\pi}^{\pi} (t^2, \sin t, \sin t) (1, \cos t, -\sin t) dt$$

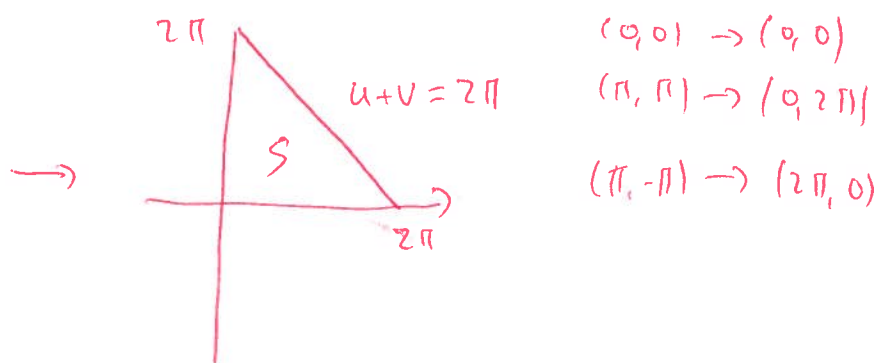
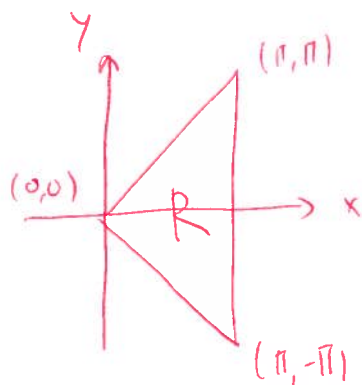
$$= \int_{-\pi}^{\pi} t^2 + \sin t \cos t - \sin^2 t dt = \left[\frac{t^3}{3} + \frac{\sin^2 t}{2} \right]_{-\pi}^{\pi} = \pi = \frac{2\pi^3}{3} - \pi$$

$$\left(\text{Note } \int_{-\pi}^{\pi} \sin^2 t dt = \frac{1}{2} \cdot 2\pi = \pi \right. \\ \left. \text{as seen in class} \right)$$

6. (15 pts) Let R be the triangle with vertices $(0,0)$, (π, π) and $(\pi, -\pi)$. Using the change of variables formula for $u = x - y$ and $v = x + y$, compute the integral $\iint_R (\cos(x-y)\sin(x+y))dA$ as an integral in du and dv .

$$x = \frac{u+v}{2} \quad y = \frac{u-v}{2}$$

$$J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{vmatrix} = -\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

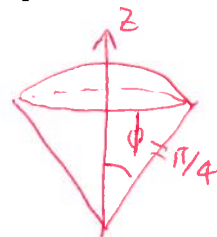


$$\int_0^{2\pi} \int_0^{2\pi-v} \cos u \sin v \cdot \frac{1}{2} du dv = \frac{1}{2} \int_0^{2\pi} \sin(2\pi-v) \sin v dv$$

$\sin(2\pi-v) = \sin(-v) = -\sin v$

$$= \frac{1}{2} \int_0^{2\pi} -\sin^2 v dv = -\pi/2$$

7. (15 pts) Compute the center of mass of the solid of unit density 1 bounded (in spherical coordinates) by $\rho = 1$ and by $0 \leq \phi \leq \frac{\pi}{4}$. (In other words this is the portion of the sphere of radius 1 with angle $\leq \pi/4$ from the z -axis.)



$$mass = \int_0^{\pi/4} \int_0^{2\pi} \int_0^1 \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi = \int_0^{\pi/4} \int_0^{2\pi} \frac{\sin \phi}{3} \, d\theta \, d\phi$$

$$= \int_0^{\pi/4} \frac{2\pi}{3} \sin \phi \, d\phi = \frac{2\pi}{3} [-\cos \phi]_0^{\pi/4} = \frac{2\pi}{3} \left[1 - \frac{\sqrt{2}}{2} \right]$$

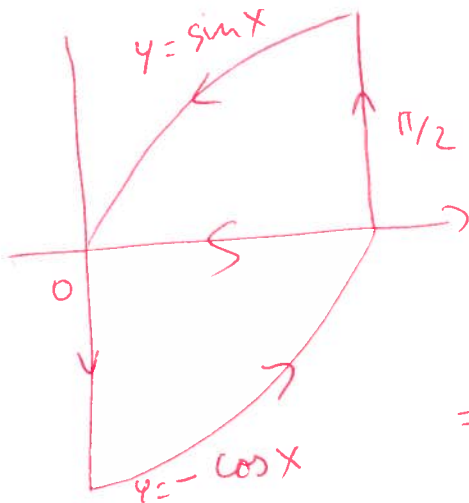
$$M_{xy} = \int_0^{\pi/4} \int_0^{2\pi} \int_0^1 \underbrace{\cos \phi}_{z} \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi = \int_0^{\pi/4} \int_0^{2\pi} \frac{\cos \phi \sin \phi}{4} \, d\theta \, d\phi = \int_0^{\pi/4} \frac{\pi}{2} \cos \phi \sin \phi \, d\phi$$

$$= \left[\frac{\pi}{2} \frac{\sin^2 \phi}{2} \right]_0^{\pi/4} = \frac{\pi}{4} \left(\frac{\sqrt{2}}{2} \right)^2 = \frac{\pi}{8}$$

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\underset{\uparrow}{0}, \underset{\uparrow}{0}, \frac{3}{16(1-\sqrt{2}/2)} \right) \quad \text{or} \quad (\bar{\rho}, \bar{\theta}, \bar{\phi}) = \left(\frac{3}{16(1-\sqrt{2}/2)}, 0, 0 \right)$$

by symmetry

8. (15 pts) Use Green's Theorem to compute $\int_C F \cdot dr$ where $F = (y^3 + x, y^5 + y)$ and C is the curve bounded below by $y = -\cos(x)$, above by $y = \sin(x)$ to the left by $x = 0$ and to the right by $x = \pi/2$. (You may find the following formulas useful: $\int_C F \cdot dr = \iint_S (N_x - M_y) dA$; and $\cos^2(x) + \sin^2(x) = 1$.)



$$\int_C F \cdot dr = - \int_0^{\pi/2} \int_{-\cos x}^{\sin x} 3y^2 dy dx$$

$$= - \int_0^{\pi/2} y^3 \Big|_{-\cos x}^{\sin x} dx = - \int_0^{\pi/2} \sin^3 x + \cos^3 x dx$$

$$= - \int_0^{\pi/2} \sin x (1 - \cos^2 x) + \cos x (1 - \sin^2 x) dx$$

$$= \left[\cos x - \sin x + \frac{\sin^3 x}{3} - \frac{\cos^3 x}{3} \right]_0^{\pi/2} =$$

$$= \left(0 - 1 + \frac{1}{3} - 0 \right) - \left(1 - 0 + 0 - \frac{1}{3} \right) = -\frac{4}{3}$$